

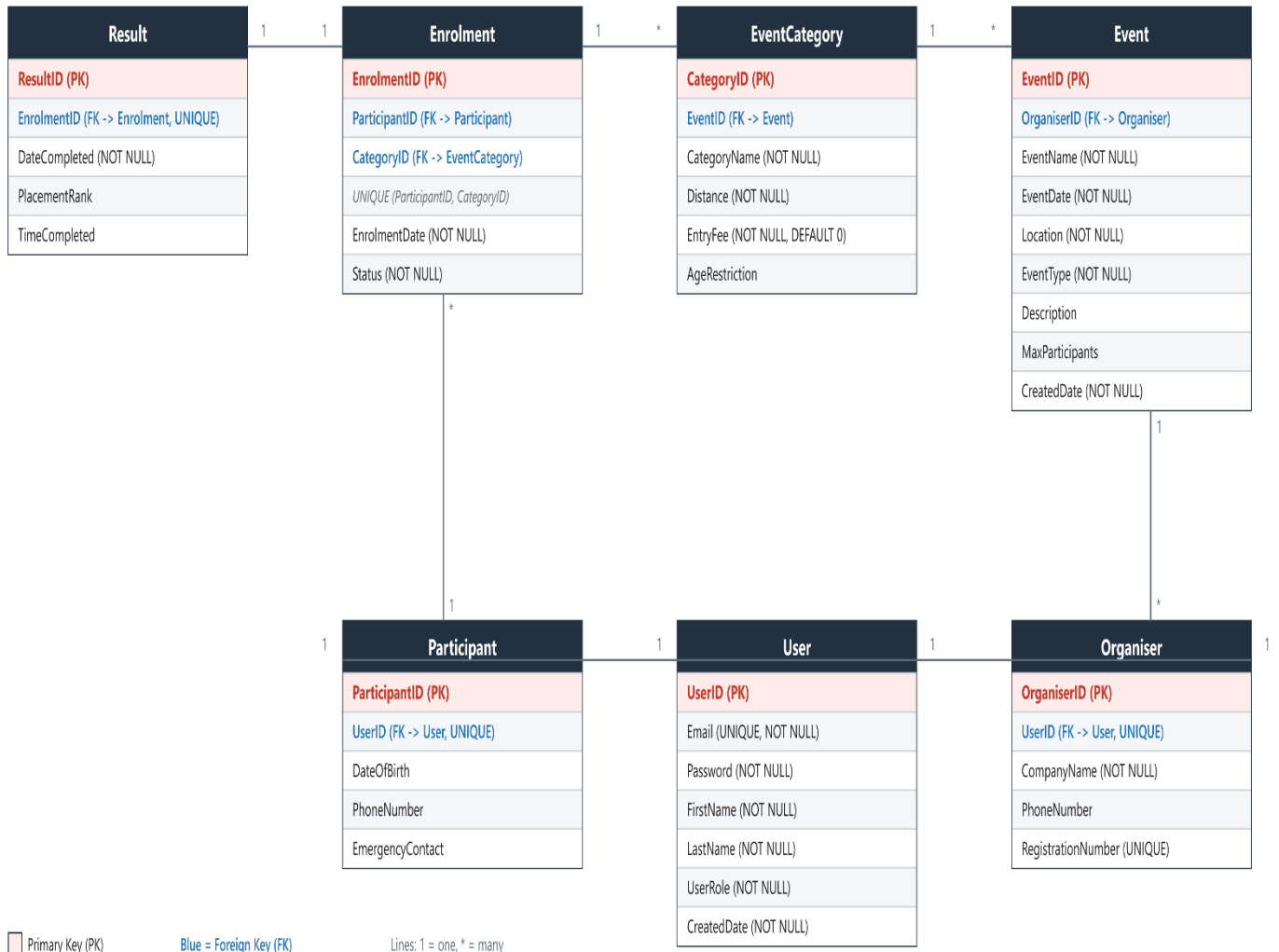
RaceDay System Documentation

Entity Relationship Diagram (ERD)

1.1 Entities (7)

- **User:** Base account table shared by both roles (Organiser and Participant).
- **Organiser (ADMIN):** 1-to-1 extension of User, for event organisers.
- **Participant:** 1-to-1 extension of User, for race entrants.
- **Event:** A race/event created by an Organiser.
- **EventCategory:** A distance/fee category within an Event (e.g., "21km").
- **Enrolment:** Junction table connecting a Participant to an EventCategory (resolves many-to-many).
- **Result:** The finishing time/rank for a completed Enrolment.

Race Event Management System - Entity Relationship Diagram



1.2 Relationships & Cardinality

Relationship	Cardinality	Notes
User → Organiser	1..1	A User row becomes an Organiser only if UserRole = "Organiser"
User → Participant	1..1	A User row becomes a Participant only if UserRole = "Participant"
Organiser → Event	1..*	One organiser can create many events
Event → EventCategory	1..*	One event has many categories (distances)
Participant → Enrolment	1..*	One participant can enrol in many categories
EventCategory → Enrolment	1..*	One category can have many participants enrolled
Enrolment → Result	1..1	Each enrolment gets at most one result, once completed

Participants and EventCategories have a many-to-many relationship. The Enrolment table resolves that connection, since a participant can join many categories, and a category can have many participants enrolled.

1.3 Design Decisions

- User is split from Organiser/Participant to avoid nullable columns and keep role-specific data (e.g. EmergencyContact vs RegistrationNumber) properly scoped.
- Enrolment is a junction table because Participant–EventCategory is many-to-many and needs its own attributes.
- Result is separate from Enrolment since not every enrolment has a result yet (only completed races).
- UNIQUE(ParticipantID, CategoryID) on Enrolment prevents duplicate registrations.
- CASCADE deletes are used down the ownership chain for data consistency.